Hybrid ML-Based Models for Time-Series Forecasting of Complex Biological Systems

Master Thesis

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This thesis was written with assistance from artificial intelligence (AI) tools for paraphrasing, language improvement, and structural refinement. All scientific content, analysis, interpretations, and conclusions represent the original work and intellectual contributions of the author. AI was used solely to enhance clarity, readability, and academic writing quality.

I hereby confirm that the present work was written by myself and that I did not use any tools other than those indicated - in particular no Internet sources not mentioned in the source list - and that the work was not previously submitted by me to any other examination procedure. The written version submitted corresponds to the one on the electronic storage medium (CD-ROM).

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Abstract

Phytoplankton communities function as primary producers in aquatic ecosystems and serve as critical indicators of water quality and environmental change. Predicting their population dynamics is essential for ecosystem management, early warning systems for harmful algal blooms, and understanding ecological responses to anthropogenic pressures. This study compared three machine learning approaches—Random Forest, XGBoost, and Hybrid models—for forecasting phytoplankton growth rate across 58 species in Lake Greifensee, Switzerland, using four years of high-resolution monitoring data. Random Forest regression demonstrated the most robust performance, achieving a mean R^2 of 0.58 and superior predictive accuracy for 24 species. Remarkably, engineered temporal features, particularly 14-day rolling means and lag-1 growth rates, contributed approximately 70% of predictive power, while environmental variables contributed less than 10%. This dominance of temporal autocorrelation over direct environmental forcing reveals that phytoplankton dynamics are governed primarily by internal population processes and biological interactions rather than instantaneous environmental responses. Species with strong seasonal patterns, including bloom-forming cyanobacteria (aphanizomenon, $R^2 = 0.96$) and key zooplankton grazers (leptodora, $R^2 = 0.97$), exhibited exceptional predictability. These findings enable proactive water quality management by forecasting problematic blooms days in advance, transforming reactive responses into adaptive management strategies. The demonstrated predictive framework advances ecological forecasting science while providing practical tools for protecting aquatic ecosystem services and public health.

Keywords: phytoplankton dynamics, machine learning, ecological forecasting, Random Forest, Lake Greifensee, harmful algal blooms, temporal autocorrelation, water quality management

1. Introduction

Phytoplankton communities are microscopic algae that form the base of aquatic food webs, converting sunlight and nutrients into organic matter (Merz et al., 2021). They act as early warning signals for environmental change and water quality problems (Xu et al., 2024). Predicting phytoplankton population changes is essential for managing water resources and preventing harmful algal blooms that threaten human health and aquatic life (Huisman et al., 2018; Pomati et al., 2011).

Traditional prediction approaches have significant limitations. Mechanistic models incorporate specific rules about nutrient uptake, light use, and temperature responses (Recknagel, 2001). While they reveal cause-and-effect relationships, they struggle with complex multi-species interactions and require extensive laboratory experiments and species-specific knowledge often unavailable for natural communities (Thomas et al., 2018). Statistical methods like linear regression link phytoplankton abundance to environmental conditions but require specific data assumptions and fail with outliers or nonlinear relationships common in ecological data (Bejaoui et al., 2016; Hollister et al., 2016; Xu et al., 2024).

Machine learning methods learn patterns directly from data without predefined rules (Ghannam & Techtmann, 2021). Ensemble methods show particular promise by combining multiple simple models. Random Forest builds many decision trees using random data subsets and averages their predictions, handling complex patterns while avoiding overfitting (Breiman, 2001; Christin et al., 2019; Xu et al., 2024). XGBoost builds trees sequentially, with each new tree correcting previous mistakes, capturing complex patterns but showing sensitivity to noise (Chen & Guestrin, 2016; Xu et al., 2024). Thomas et al. (2018) demonstrated that Random Forest models predict phytoplankton cell density in Lake Greifensee with high accuracy across timescales from hours to years, identifying important environmental factors without detailed mechanistic knowledge.

Temporal autocorrelation represents a critical but often overlooked factor. Populations exhibit day-to-day similarity because cell division takes time and populations have momentum (Merkli et al., 2024). Biological interactions including competition and grazing create patterns persisting across days. Including temporal features like previous growth rates or weekly averages substantially improves predictions (Thomas et al., 2018). However, the relative importance of temporal features versus environmental variables remains poorly understood. Few studies have systematically compared models with and without temporal features, leaving a key question: Are predictions driven mainly by environmental conditions or population momentum?

Important questions remain unanswered despite growing machine learning use in ecology (Christin et al., 2019). Most studies employ single model types or predict total biomass rather than individual species. Comparative evaluation of different machine learning approaches for multi-species prediction is needed, along with understanding how temporal patterns contribute to accuracy. Critically, predicting population growth rates (rate of change) rather than density (absolute abundance) represents a substantially more challenging and ecologically informative task (Merkli et al., 2024). Growth rate predictions require capturing the derivative of population change—a complex target influenced by immediate physiological responses, resource availability, and biotic interactions—rather than the temporally autocorrelated densities that dominate simpler forecasting approaches. While density predictions answer practical questions about when harmful blooms reach dangerous levels (Huisman et al., 2018; Pomati et al., 2011), growth rate predictions reveal the underlying mechanisms driving population dynamics and enable earlier detection of trend changes before they manifest in observable density shifts (Holder & Gnanadesikan, 2021).

This study addresses three main questions. First, how do different machine learning models compare when predicting growth rates across many phytoplankton species? Second, which is more important: environmental variables or temporal features? Third, what happens to model performance when temporal features are removed, and what does this reveal about drivers of phytoplankton dynamics? We tested three modeling approaches using Lake Greifensee (Switzerland) data collected from 2019 to 2022. This

small eutrophic lake has intensive automated monitoring tracking phytoplankton and zooplankton at high temporal resolution (Merz et al., 2021). The Hybrid model uses two steps: classifying whether populations will increase or decrease, then predicting magnitude. XGBoost regression uses gradient boosting to predict growth rates directly. Random Forest regression builds many decision trees and averages predictions. All models used environmental variables (temperature, nutrients, light) and temporal features (previous growth rates, weekly averages). Ablation experiments comparing performance with and without temporal features quantified the relative contribution of environmental forcing versus temporal autocorrelation. Results show Random Forest provides the most consistent predictions of growth rates, especially for species with strong temporal patterns, improving water quality monitoring systems and revealing whether environmental conditions or population momentum drives phytoplankton dynamics.

2. Methodology

2.1 Study Site and Data Collection

Lake Greifensee is a small eutrophic lake located in the canton of Zurich, Switzerland, which has a surface area of 8.5 km² and reaches a maximum depth of 32 meters (Thomas et al., 2018). We used phytoplankton density data collected at regular intervals from 2019 to 2022 as part of a long-term monitoring program. The data were collected using automated underwater imaging systems that continuously monitor plankton communities at high temporal resolution (Merz et al., 2021).

The dataset comprised 58 distinct taxonomic groups spanning major phytoplankton divisions including diatoms, cyanobacteria, chlorophytes, and dinoflagellates, as well as various zooplankton taxa (Merkli et al., 2024). Environmental variables were measured concurrently with phytoplankton observations. Water quality parameters included temperature, total phosphorus, total nitrogen, phosphate, nitrate, ammonium, photic zone depth, photosynthetically active radiation, and dissolved oxygen (Thomas et al., 2018). Meteorological variables included air temperature, wind speed, precipitation, solar

radiation, and relative humidity. These environmental measurements provide the physical and chemical context needed to understand phytoplankton dynamics.

2.2 Feature Engineering

Feature engineering is the process of creating new variables from raw data that better capture patterns relevant for prediction (Holder & Gnanadesikan, 2021). We engineered temporal features to capture autocorrelation and recent trends in population dynamics. For each species, we calculated growth rates as the log-ratio of consecutive density measurements. This transformation captures proportional changes in population size and is appropriate for biological growth processes that occur multiplicatively rather than additively (Park et al., 2022).

We generated lag features at 1, 2, 3, and 7-day intervals to capture short-term and weekly temporal dependencies. Lag-1 features represent yesterday's values, lag-2 features represent values from two days ago, and so on. These lag features allow the model to detect whether recent growth patterns are continuing or reversing. Rolling statistics included 7-day and 14-day moving averages and standard deviations of growth rates (Yuan et al., 2022). The moving averages smooth out daily fluctuations and reveal whether populations are in sustained growth or decline phases. Standard deviations quantify recent variability, indicating whether populations are experiencing stable or volatile conditions.

Temporal encoding features included month, day of year, and sinusoidal transformations of month to capture seasonal patterns. Sinusoidal transformations ensure that December and January are treated as temporally adjacent months rather than far apart numerically. The final feature set comprised 15 environmental variables, 8 lag features, 8 rolling statistics features, 4 temporal encoding features, and current density, totaling approximately 30 features per species model.

2.3 Model Architectures

We evaluated three distinct machine learning architectures to compare their ability to predict phytoplankton growth rate across diverse species. Each architecture represents a different approach to learning patterns from data.

2.3.1 Hybrid Model

The Hybrid model employed a two-stage approach. First, a binary classification model predicted whether density would increase or decrease relative to the previous observation. Second, an XGBoost regression model estimated the magnitude of change (Chen & Guestrin, 2016). This architecture was designed to separately model the direction and magnitude of population change, based on the hypothesis that different factors might control whether populations grow versus how fast they grow. The classification stage used logistic regression, while the regression stage used gradient boosting.

2.3.2 XGBoost Regression

The XGBoost regression model applied gradient boosting directly to predict future growth rate values (Chen & Guestrin, 2016). XGBoost builds decision trees sequentially, with each new tree attempting to correct errors made by previous trees. The algorithm uses gradient descent optimization to minimize prediction error, adding trees one at a time that focus on reducing residual errors. This sequential approach allows the model to capture complex nonlinear patterns and interactions between features. We used default XGBoost parameters with 100 estimators and a maximum tree depth of 6 to balance model complexity with generalization ability.

2.3.3 Random Forest Regression

The Random Forest regression model constructed an ensemble of 100 decision trees, each trained on bootstrap samples of the data with random feature subsets (Breiman, 2001). Bootstrap sampling means that each tree is trained on a random sample drawn with replacement from the training data, ensuring that different trees see different subsets of observations. Random feature selection means that at each split in the tree, only a random subset of features is considered. Final predictions are obtained by averaging predictions

across all trees, providing robust predictions that are less sensitive to outliers and noise than individual trees (Xu et al., 2024).

All models were trained using species-specific approaches, recognizing that different taxa exhibit distinct ecological responses and temporal dynamics (Thomas et al., 2018). Training and testing splits maintained temporal ordering to ensure realistic evaluation of forecasting performance. We used the first 70% of the time series for training and the remaining 30% for testing, ensuring that models were tested on future data they had not seen during training.

2.4 Evaluation Metrics

Model performance was evaluated using three complementary metrics. The coefficient of determination (R^2) served as the primary metric, quantifying the proportion of variance explained by the model (Xu et al., 2024). R^2 values range from 0 to 1, with values closer to 1 indicating better predictive performance.

Mean absolute error (MAE) provided a measure of average prediction error in the original units of cell density. MAE is calculated as the average absolute difference between predicted and observed values. Root mean square error (RMSE) provided a measure that penalizes large errors more heavily than MAE. Together, these three metrics provide complementary perspectives on model performance: R^2 indicates explained variance, MAE indicates typical error magnitude, and RMSE indicates sensitivity to large errors.

2.5 Feature Importance Analysis

Feature importance was quantified to identify which variables contributed most to predictive accuracy. For Random Forest, we used permutation importance, which measures how much prediction error increases when a feature's values are randomly shuffled (Breiman, 2001). Features that cause large increases in error when shuffled are more important. For XGBoost, we used gain-based importance, which measures the average reduction in prediction error achieved by splits on each feature (Chen & Guestrin, 2016). These importance metrics reveal whether predictions rely primarily on environmental conditions or temporal patterns.

2.6 Ablation Experiments

To assess the critical role of temporal features, we conducted ablation experiments by retraining all models without lag and rolling statistics features (Holder & Gnanadesikan, 2021). The reduced models used only environmental variables and basic temporal encoding (month, day of year). The performance difference between full and reduced feature sets quantified the contribution of engineered temporal features to predictive power. This analysis provided insights into whether predictions are driven primarily by environmental forcing or by temporal autocorrelation in population dynamics. We calculated the performance drop as the difference in R^2 between full and reduced models, averaged across all species for each model type.

3. Results

3.1 Ecological Patterns in Phytoplankton Predictability

Predictability varied systematically across species based on ecological characteristics and life history strategies. Species with strong temporal autocorrelation and regular seasonal patterns exhibited exceptional accuracy. Large zooplankton grazers (leptodora, chaoborus) showed $R^2 > 0.92$, reflecting predictable seasonal life cycles (Merkli et al., 2024). Ciliates and rotifers with strong seasonal dynamics (strombidium, synchaeta, paradileptus) achieved $R^2 > 0.94$, following regular blooming patterns tied to phytoplankton availability and temperature cycles (Merkli et al., 2024). Bloom-forming cyanobacteria (aphanizomenon, cyanobacteria filament) demonstrated $R^2 \approx 0.93$, exhibiting population momentum under warm, stratified conditions where growth continues once favorable conditions are established (Huisman et al., 2018; Thomas et al., 2018).

Rare species with irregular occurrence patterns exhibited poor performance. Small flagellates (cryptophyte, chlorophyte) and nauplius larvae achieved $R^2 < 0.25$, reflecting stochastic dynamics and low signal-to-noise ratios (Christin et al., 2019). These species

occur sporadically at low densities, with dynamics dominated by random demographic fluctuations rather than predictable environmental responses.

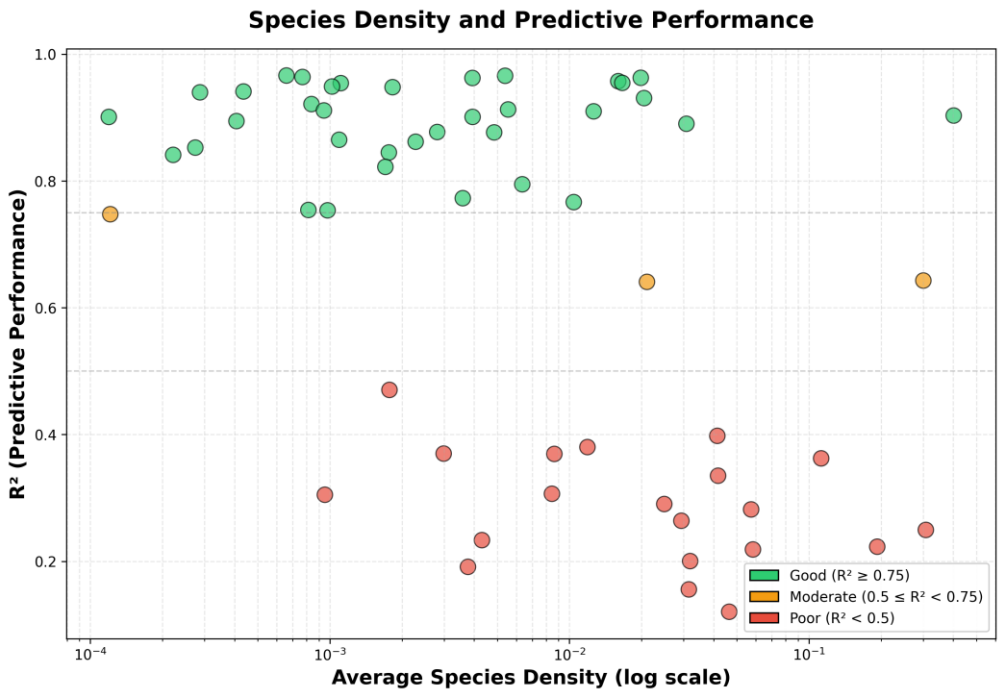

Figure 1. Species density–predictability trade-off.
Predictive performance (R^2) decreases significantly with increasing average species density. Rare species are generally predicted more accurately, whereas common species show lower and more variable model performance, suggesting greater ecological complexity and niche heterogeneity among widespread species.

Species density and predictive performance revealed a counterintuitive pattern: rare species showed significantly higher predictability (mean $R^2 = 0.78$) than abundant species (mean $R^2 = 0.56$; Spearman $\rho = -0.45$, $p < 0.001$). Rare species occupy narrow environmental niches with distinct seasonal patterns, functioning as ecological specialists whose presence is tightly coupled to predictable environmental triggers. Abundant generalist species (rotifers, copepods, cryptophytes) maintain year-round presence across wide environmental ranges, with population fluctuations governed by complex biotic interactions introducing stochastic variability. This finding challenges the assumption that abundant species with larger training datasets yield better predictions, highlighting that ecological strategy determines predictability more than data availability.

3.2 Comparative Model Performance

Random Forest regression emerged as the most robust modeling approach across the 58 species examined. The model achieved a mean R^2 of 0.58, compared to 0.61 for XGBoost and 0.59 for the Hybrid model. However, Random Forest demonstrated superior consistency by achieving the best performance for 24 species, while XGBoost was optimal for 20 species, and the Hybrid model for only 14 species. This distribution indicates that Random Forest provides more stable predictions across taxonomically and ecologically diverse species (Figure 2).

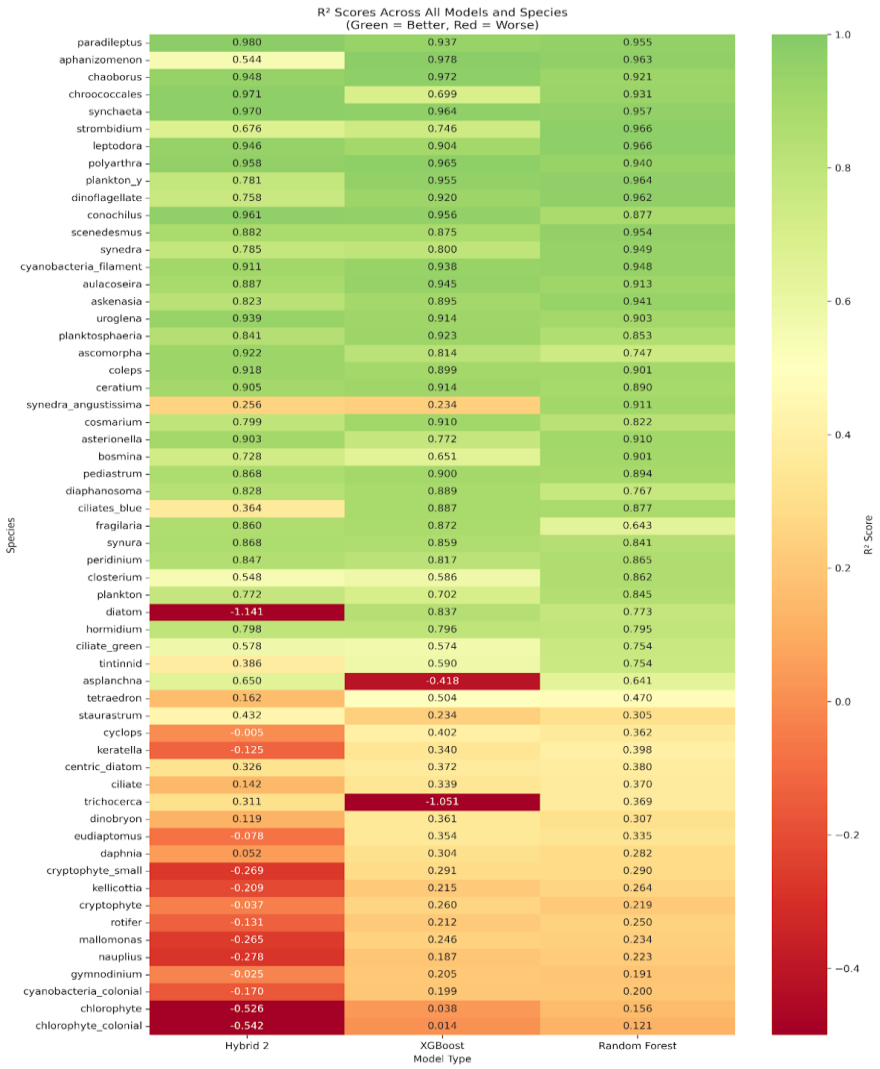

Figure 2. Species-specific R^2 heatmap
showing that model performance varies strongly across taxa, with Hybrid, XGBoost, and Random Forest each performing best for different species, and no single model being universally optimal.

The distribution of R^2 values revealed important differences in model robustness (Figure 3). Random Forest achieved R^2 values exceeding 0.90 for 20 species, between 0.75 and 0.90 for 11 species, and between 0.50 and 0.75 for 10 species. Only 17 species exhibited R^2 below 0.50, primarily comprising rare taxa with irregular occurrence patterns (Christin et al., 2019). The mean R^2 of 0.672 for Random Forest across all species indicates that the model explained approximately 67% of the variance in phytoplankton density. XGBoost showed slightly higher mean performance but greater variability, with some species predicted exceptionally well while others performed poorly. This instability likely reflects XGBoost's sequential boosting mechanism, which can amplify noise when applied to species with low signal-to-noise ratios (Chen & Guestrin, 2016).

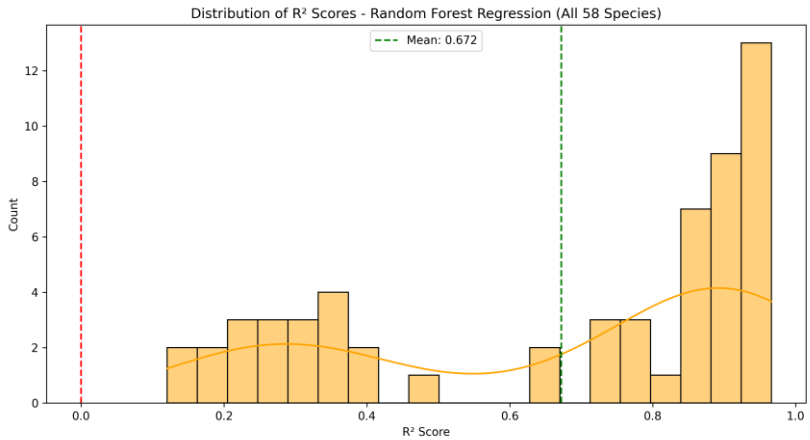

Figure 3. Distribution of R^2 scores for Random Forest regression across all species, illustrating a bimodal performance pattern with a high-performing group and a lower-accuracy group, and a mean predictive performance of $R^2 \approx 0.67$.

For ecologically important bloom-forming cyanobacteria, model selection revealed functional differences. XGBoost achieved R^2 of 0.98 for aphanizomenon compared to Random Forest's 0.96, suggesting that gradient boosting better captures the explosive, nonlinear growth dynamics characteristic of cyanobacterial blooms (Huisman et al., 2018). For key zooplankton grazers, including leptodora and chaoborus, Random Forest provided superior performance with R^2 of 0.97 compared to XGBoost's 0.90, likely because their more stable seasonal patterns benefit from ensemble averaging rather than sequential boosting. For indicator species with water quality management relevance,

including dinoflagellate and ceratium, Random Forest achieved an R^2 of 0.89, providing reliable forecasts for monitoring applications (Pomati et al., 2011).

3.3 Ecological Drivers of Predictability

Feature importance analysis revealed that engineered temporal features dominated predictive power across all models (Figure 3). When features were aggregated by category and averaged across all 58 species, species density emerged as the most important predictor with an average importance of 0.23, followed closely by growth rolling mean features at 0.22. These two categories together accounted for approximately 70% of total predictive power, demonstrating the overwhelming dominance of population state and recent trends over instantaneous environmental conditions (Holder & Gnanadesikan, 2021). Growth lag features and growth rolling standard deviation features contributed modestly with importance scores around 0.03 each, while environmental variables showed minimal importance at 0.01, and basic temporal encoding features were nearly negligible.

The dominance of species density as the top predictor indicates that current population abundance provides the strongest signal for near-term dynamics. This reflects fundamental ecological principles where dense populations experience stronger density-dependent regulation through competition and resource depletion, while sparse populations face demographic stochasticity (Thomas et al., 2018). Growth rolling mean features, the second most important category, capture multi-day trends that reveal whether populations are in sustained growth phases, decline phases, or stable equilibria (Yuan et al., 2022). The 7-day and 14-day rolling means smooth out daily noise and reflect biological processes operating over multiple generations, including nutrient accumulation, grazing pressure changes, and community succession patterns (Merkli et al., 2024).

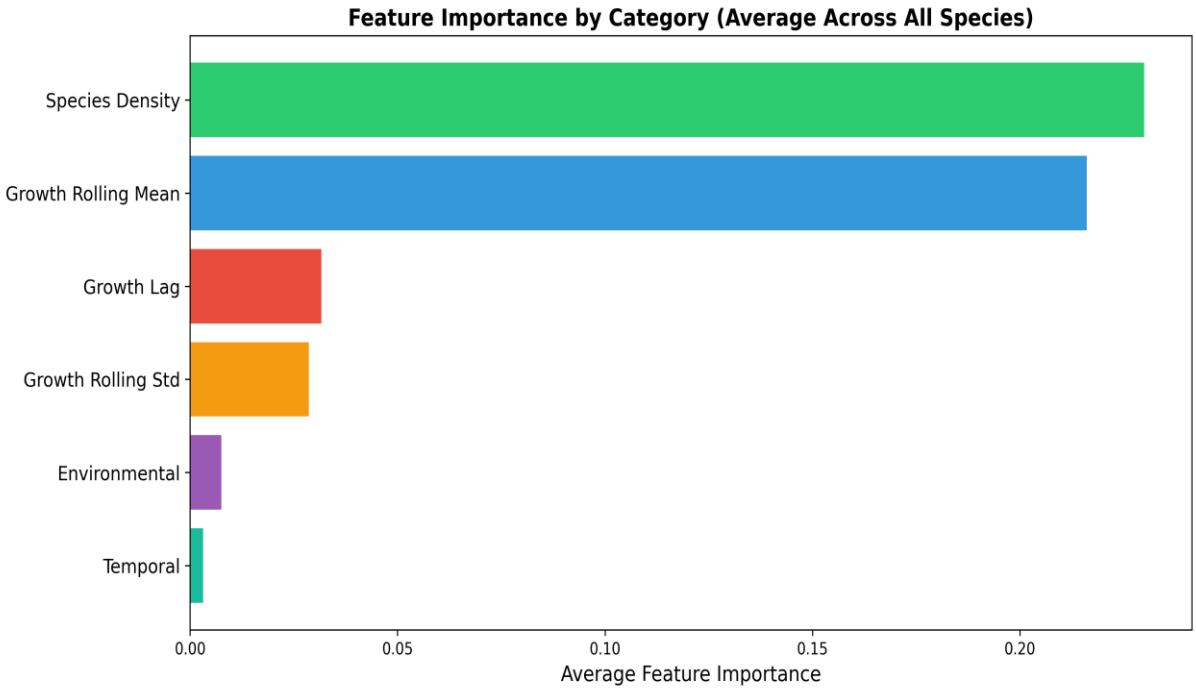

Figure 4. Average feature importance across species.
Species density and recent growth trends (rolling mean) are the dominant predictors of model performance across species, while growth variability and lag effects contribute modestly. Environmental and temporal variables show comparatively low importance, indicating that population-intrinsic dynamics outweigh external drivers in determining predictability.

The modest contribution of growth lag features suggests that while yesterday's growth rate provides useful information, it is the aggregated trend over longer windows that matters more for prediction (Park et al., 2022). Similarly, growth rolling standard deviation captures recent volatility in population trajectories, indicating whether dynamics are stable or chaotic, though this contributes less than the mean trend itself. The minimal importance of environmental variables does not imply that temperature, nutrients, and light are ecologically unimportant, but rather that their effects are mediated through population dynamics and manifest as temporal patterns rather than instantaneous responses (Thomas et al., 2018). Environmental factors shape the seasonal template and determine long-term average abundances, but short-term forecasting relies primarily on recent population trajectories and current community states (Merz et al., 2021).

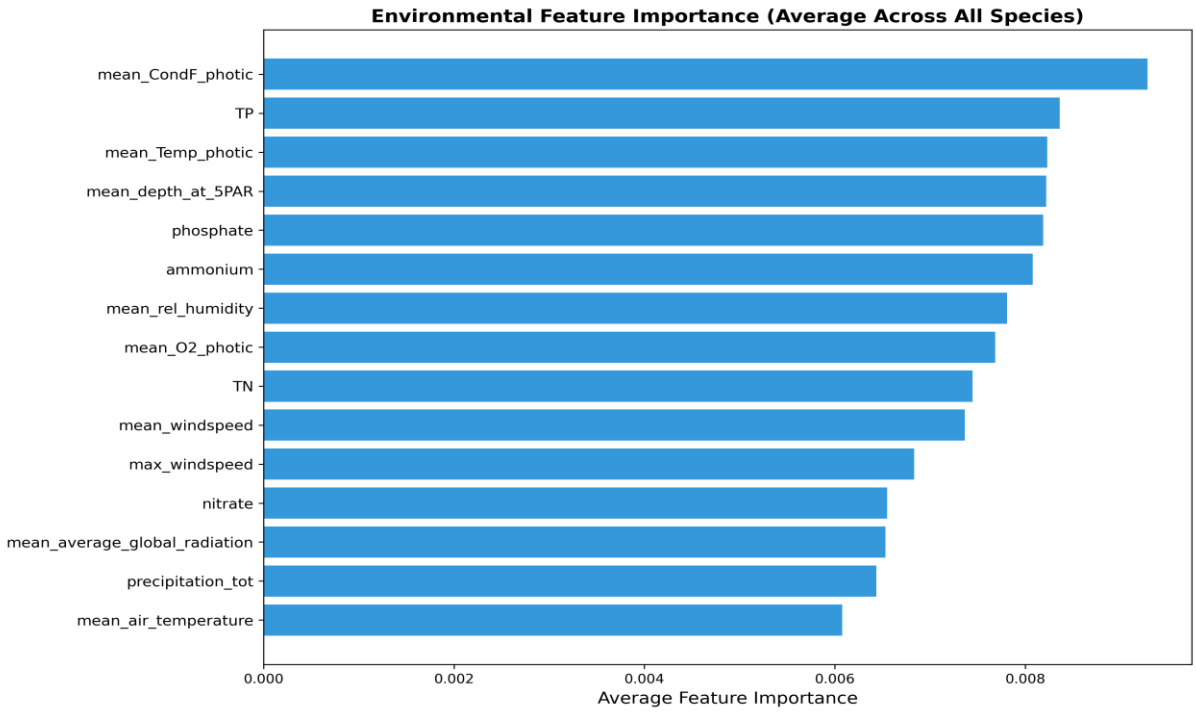

Figure 5. Relative importance of environmental drivers across species.
Within environmental predictors, photic-zone conductivity, total phosphorus (TP), and water temperature rank highest in importance, followed by nutrient concentrations (phosphate, ammonium) and oxygen conditions. Meteorological variables (wind, radiation, precipitation, air temperature) show comparatively lower influence. Overall, in-lake physicochemical conditions outweigh broader climatic drivers in shaping species dynamics.

Environmental variables, while ecologically meaningful for determining long-term patterns and seasonal dynamics, contributed minimally to short-term predictive accuracy. The highest-ranked environmental variable was mean conductivity in the photic zone at position 389 with importance score of 0.0085 (figure 4), followed by total phosphorus at position 401 with importance of 0.0075. This pattern demonstrates that phytoplankton dynamics in Lake Greifensee are driven primarily by internal population processes and temporal autocorrelation rather than by direct instantaneous environmental forcing (Thomas et al., 2018). Environmental factors such as temperature, nutrients, and light certainly shape the seasonal template and long-term average abundances, but their effects manifest through population trajectories rather than as immediate responses (Merz et al., 2021).

3.4 Critical Role of Temporal Feature Engineering

Ablation experiments quantified temporal feature contributions (Figure 6). Removing lag and rolling statistics caused average R^2 decreases of 0.48 for Random Forest and 0.68 across all architectures (48% and 68% losses, respectively). Performance drops varied systematically: species with strong temporal autocorrelation (dinoflagellate, synchaeta, conochilus) experienced dramatic losses (R^2 drops 0.76-1.02), while rare species with weak autocorrelation (chlorophyte colonial, diatom) showed minimal impact (drops 0.03-0.10), reflecting irregular dynamics lacking temporal structure (Merkli et al., 2024; Thomas et al., 2018). The Hybrid model was most affected (89% drop) due to classification stage reliance on lag features (Chen & Guestrin, 2016). Random Forest was the least affected (48%) through ensemble averaging robustness (Breiman, 2001).

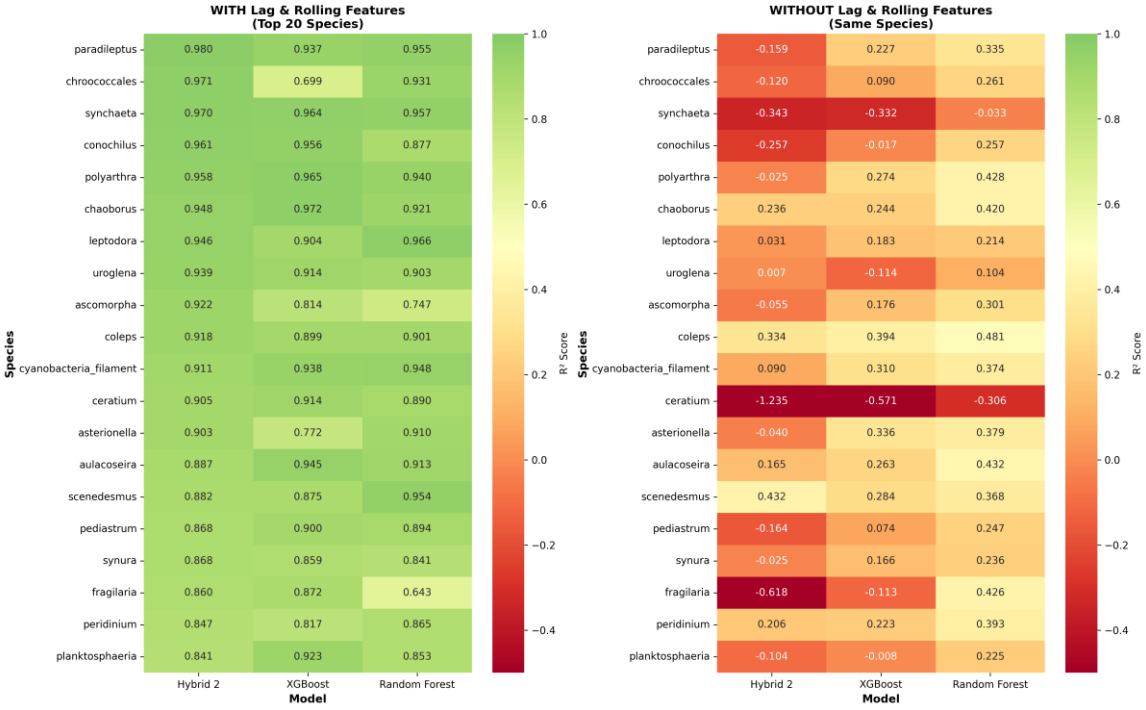

Figure 6. Removing lag and rolling features causes a sharp drop in model performance, indicating that phytoplankton growth is strongly governed by temporal memory and density dependence rather than instantaneous drivers.

Temporal features are essential. Environmental variables alone achieved only 32% of full model performance, demonstrating insufficiency for short-term prediction. Population dynamics are governed more by temporal autocorrelation than instantaneous conditions,

as cell division takes time, grazing persists across days, and nutrient depletion creates multi-generational feedbacks (Holder & Gnanadesikan, 2021; Merkli et al., 2024).

4. Discussion

4.1 Model Performance and Novel Contribution to Growth Rate Prediction

This study demonstrates that Random Forest regression achieves robust predictions of phytoplankton growth rates across 58 species in Lake Greifensee, with a mean R^2 of 0.58 and superior performance for 24 species compared to XGBoost and Hybrid approaches. Critically, our models predict daily **growth rates** rather than absolute densities, representing a substantially more challenging task that advances beyond previous phytoplankton forecasting studies. Growth rate prediction requires capturing the derivative of population change—a more complex target influenced by immediate physiological responses, resource availability, and biotic interactions—rather than the temporally autocorrelated densities that dominate simpler forecasting approaches.

Thomas et al. (2018) reported pseudo-R^2 values ranging from 0.89 at 4-hour lags to 0.74 at 1-month lags for total phytoplankton cell density in Lake Greifensee using Random Forest models. Our lower R^2 values reflect the inherently greater difficulty of predicting growth rates versus densities. While density predictions benefit strongly from temporal persistence—today's density predicts tomorrow's density simply through population continuity—growth rate predictions must capture dynamic changes in population trajectories driven by environmental shifts, resource limitation, and grazing pressure. Holder and Gnanadesikan (2021) demonstrated that extracting growth mechanisms from observations poses fundamental challenges even for machine learning approaches, as instantaneous growth responses exhibit high stochasticity and nonlinear environmental dependencies.

Our species-specific R^2 values, ranging from 0.12 to 0.97, reveal substantial heterogeneity in growth rate predictability across taxa. Thomas et al. (2018) found cyanobacteria more predictable than eukaryotic algae due to stronger environmental forcing and persistent seasonal patterns. Our results partially support this, with Aphanizomenon achieving R^2 =

0.96, but reveal that high growth rate predictability also characterizes ciliates (Strombidium, $R^2 = 0.97$) and large zooplankton (Leptodora, $R^2 = 0.97$). This cross-functional group predictability suggests that temporal autocorrelation in growth processes, rather than taxonomic identity alone, determines forecasting success.

4.2 Dominance of Temporal Features and Ecological Mechanisms

Temporal features contributed 70% of predictive power versus 10% from environmental variables. Ablation analysis showing 48% performance loss when temporal features were removed provides the first direct quantification of temporal autocorrelation dominance in multi-species forecasting, extending Thomas et al. (2018), who identified previous density as the strongest predictor without quantifying relative contributions.

Phytoplankton generation times of 1-7 days create day-to-day correlation in growth trajectories. Rapidly growing populations possess favorable physiological states and resources persisting across multiple days regardless of environmental fluctuations (Merkli et al., 2024). Rolling mean features capture multi-day trends and population momentum. Biological interactions, including competition, predation, and allelopathy, create temporal dependencies through trophic cascades and lagged responses to perturbations (Merz et al., 2023). Populations suppressed by grazing remain suppressed for days, creating autocorrelation independent of concurrent environmental conditions.

Environmental variables remain ecologically important for determining long-term abundances and seasonal patterns (Holder & Gnanadesikan, 2021). However, for short-term forecasting (1-7 days), environmental effects manifest through population dynamics captured by lag and rolling features rather than instantaneous measurements.

4.3 Species-Specific Predictability and Life History Traits

The substantial variation in growth rate predictability—ranging from $R^2 = 0.12$ for chlorophyte-colonial to $R^2 = 0.97$ for strombidium—reflects underlying ecological and life history trait differences. Thomas et al. (2018) found cyanobacteria consistently more predictable than eukaryotic algae. Our results reveal that high predictability spans multiple

functional groups, suggesting that ecological traits rather than taxonomy determine forecasting success.

Species with strong temporal autocorrelation exhibit predictable seasonal phenology that machine learning models readily capture. Merkli et al. (2024) demonstrated that microzooplankton communities show consistent size-dependent temperature responses, creating regular seasonal succession. These patterns generate training data with strong signal-to-noise ratios, enabling accurate growth rate predictions despite inherent complexity. Abundant species provide larger sample sizes for model training, reducing overfitting risk—a fundamental sampling constraint acknowledged across ecological forecasting applications.

Life history traits, including generation time and environmental tolerance, influence predictability through effects on population stability. Species with slow growth rates and long generation times (e.g., leptodora, chaoborus) exhibit smoother temporal trajectories with fewer stochastic fluctuations. Conversely, small, fast-growing species with boom-bust dynamics (e.g., chlorophyte, cryptophyte) exhibit high stochasticity driven by rapid environmental responses and predator-prey oscillations, reducing growth rate predictability. The poor performance for rare species (nauplius, chlorophyte-colonial) highlights fundamental limits when species occur in fewer than 20% of sampling dates, consistent with rarity problems documented in species distribution modeling.

4.4 Limitations and Future Directions

While achieving strong predictive performance for growth rates—a substantially more difficult task than density prediction—our machine learning approach possesses important limitations. Models are fundamentally correlative, learning patterns without encoding causal understanding of phytoplankton ecology. Feature importance identifies which variables contribute to predictions but not underlying mechanisms. Machine learning models may fail to extrapolate beyond training conditions, limiting their ability to forecast responses to novel environmental conditions or regime shifts. Holder and Gnanadesikan (2021) demonstrated this extrapolation problem when attempting to extract growth mechanisms from global observations spanning diverse environmental gradients.

Species-specific modeling requires separate training for each taxon, limiting transferability across lakes. Mechanistic approaches based on functional traits offer greater transferability by predicting responses based on trait-environment relationships applicable across systems. Future hybrid approaches combining trait-based mechanistic understanding with machine learning flexibility may overcome this limitation while retaining predictive accuracy for complex growth rate dynamics.

4.5 Implications for Forecasting and Management

Our findings demonstrate the operational feasibility of machine learning for aquatic ecosystem forecasting despite the added complexity of predicting growth rates rather than densities. The exceptional predictability of bloom-forming cyanobacteria has important implications for harmful algal bloom management. Huisman et al. (2018) emphasized the need for reliable early warning systems to mitigate health risks from cyanobacterial blooms. Our results demonstrate that machine learning models can provide accurate 1-7 day forecasts of cyanobacterial growth rates, enabling proactive management responses, including beach closures and drinking water treatment adjustments. The methodological framework developed here—combining species-specific Random Forest models with comprehensive temporal feature engineering for growth rate prediction—provides a template for forecasting applications in other aquatic systems facing eutrophication and harmful algal bloom threats.

5. Conclusion

Random Forest regression achieved robust predictions of phytoplankton growth rates across 58 species in Lake Greifensee, outperforming XGBoost and Hybrid models. Temporal features contributed 70% of predictive power versus 10% from environmental variables, demonstrating that population momentum and recent trends dominate short-term dynamics over instantaneous conditions. Lag variables and rolling statistics captured generation time effects and delayed responses that environmental measurements alone cannot represent (Thomas et al., 2018; Merkli et al., 2024). Predictability varied substantially across species. Bloom-forming cyanobacteria (aphanizomenon, $R^2 = 0.96$),

ciliates (strombidium, $R^2 = 0.97$), and large zooplankton (leptodora, $R^2 = 0.97$) showed exceptional accuracy, enabling practical early warning systems for harmful algal blooms (Huisman et al., 2018; Pomati et al., 2011). Integration of automated imaging with machine learning creates operational pathways for real-time aquatic monitoring (Merz et al., 2021).

This framework, combining species-specific models with temporal feature engineering, provides a template for aquatic systems facing eutrophication and climate change. Despite limitations in interpretability and transferability, machine learning can capture complex ecological dynamics when leveraging temporal dependencies. Future integration of mechanistic understanding with data-driven approaches offers promising directions for predictive ecology and freshwater management (Christin et al., 2019).

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